Design and Simulation of a 6-Bit X-Band CMOS Phase Shifter in a 180-nm CMOS Process

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***Abstract*—This paper documents the design and reproducible postprocessing of a 6-bit digital phase shifter for X-band operation in a TSMC 180-nm CMOS design environment. Six binary-weighted cells with nominal increments of 5.625, 11.25, 22.5, 45, 90, and 180 degrees are cascaded to provide 64 control states. The circuit combines switched low-pass, band-pass, and high-pass networks synthesized for a 10-GHz center frequency and a 50-ohm system. Historically reported NMOS samples give an equivalent on-state resistance and off-state capacitance of approximately 3.33 ohms and 57.8 fF at 10 GHz, respectively; the native device-characterization export is unavailable. Preserved Cadence S-parameter exports are independently processed for all 64 states at 9, 10, and 11 GHz. The offset-removed phase-error root-mean-square value is 2.28 degrees at 10 GHz and 3.67 degrees over all 192 state-frequency samples. Across the aggregate data, the mean insertion loss is 15.69 dB with 1.73 dB variation, while the mean and minimum input return losses are 9.49 and 5.48 dB, respectively. The results confirm accurate digital phase progression but identify transmission loss and impedance matching as the dominant limitations. The artifact includes the preserved OpenAccess schematic source and top-level layout and reproduces the numerical analysis, but not the original Cadence simulation environment, post-layout verification, or measured performance.**

***Index Terms*—CMOS phase shifter, digital phase shifter, X-band, S-parameters, phased array, 180-nm CMOS.**

I. Introduction

Digitally stepped phase shifters are key radio-frequency (RF) blocks in phased-array transmit/receive modules because they set the relative phase at each antenna element and therefore control the array beam direction. A 6-bit architecture provides 64 discrete phase states and a nominal least significant bit (LSB) of 5.625 degrees. Switched high-pass/low-pass and related filter networks are established passive implementations for X-band phase control [1], [3]. CMOS integration can simplify digital control and reduce static power, but the finite switch resistance, off-state capacitance, substrate loss, and limited passive-element quality factor make low-loss operation difficult at 8–12 GHz [2], [4].

This work documents a 6-bit X-band design centered at 10 GHz and evaluates its complete 64-state response. The technical objectives are fourfold: describe the binary-weighted circuit architecture, quantify the switch nonidealities used in design tuning, define a reproducible data-reduction method, and separate demonstrated simulation results from unverified physical-design claims. Particular attention is given to phase-error definitions because a common frequency-dependent phase offset can otherwise be confused with state-dependent phase nonlinearity.

II. Circuit Design

A. Digital Phase Coding

Let $b_k \in \{0, 1\}$ denote the control of phase bit k, with $k = 0$ for the 5.625-degree LSB and $k = 5$ for the 180-degree most significant bit. The ideal relative phase of state i is

$$\phi_{i,\mathrm{ideal}} = 5.625^\circ \sum_{k=0}^{5} 2^k b_k, \qquad i = \sum_{k=0}^{5} 2^k b_k. \tag{1}$$

Thus, the 64 codes cover 0 to 354.375 degrees in 5.625-degree increments. Each cell switches between a reference path and a phase-shifting path. The six cells are cascaded between 50-ohm ports, as shown in Fig. 1. Gate controls use nominal levels of 0 and 1.8 V, and large gate resistors isolate the RF path from the control source.

B. NMOS Switch Characterization

The switch model uses an NMOS with channel length $L = 0.18\ \mu\mathrm{m}$, total gate width $W = 64\ \mu\mathrm{m}$, and 16 fingers. Its source and drain are characterized as a two-port network in a 50-ohm environment, the body is grounded, and the gate is biased through a 20-kΩ resistor. Equivalent on-state resistance and off-state capacitance are estimated from converted two-port parameters as

$$\begin{aligned} R_{\mathrm{on}}(f) &\approx \mathrm{Re}\{Z_{11}(f)\}\big|_{V_G = 1.8\,\mathrm{V}}, \\ C_{\mathrm{off}}(f) &\approx \left.\frac{\mathrm{Im}\{Y_{11}(f)\}}{2\pi f}\right|_{V_G = 0\,\mathrm{V}}. \end{aligned} \tag{2}$$

The original report lists R_{on} values of 3.61, 3.33, and 3.11 Ω at 9, 10, and 11 GHz, respectively. At 10 GHz, $\mathrm{Im}\{Y_{11}\} = 3.6317$ mS gives $C_{\mathrm{off}} \approx 57.8$ fF and an off-state reactance magnitude of approximately 275 Ω. The three reported Z_{11} and Y_{11} samples are transcribed in the repository, and their arithmetic is reproducible. However, the native NMOS sweep, simulator state, process corner, and PDK revision are unavailable, so the underlying device simulation is not reproducible. The reported parasitics are not negligible relative to the intended lumped elements, particularly for the small phase bits.

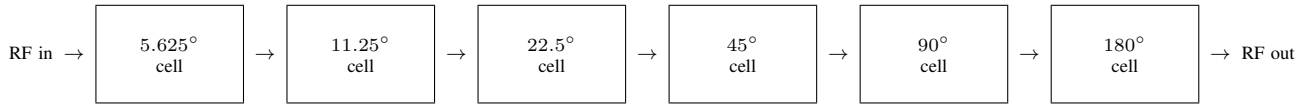


Fig. 1. Functional architecture of the cascaded 6-bit phase shifter. Each binary control selects the reference or phase-shifting path of one cell.

TABLE I
FIRST-ORDER IDEAL SYNTHESIS VALUES AT 10 GHZ

Phase bit	Analytical starting values
5.625°	$L_1 = 78.19$ pH, $C_1 = 3.240$ pF; $L_2 = 39.09$ pH, $C_2 = 31.20$ fF
11.25°	$L_1 = 156.8$ pH, $C_1 = 1.616$ pF; $L_2 = 78.38$ pH, $C_2 = 62.10$ fF
22.5°	$L_1 = 316.6$ pH, $C_1 = 0.800$ pF; $L_2 = 158.3$ pH, $C_2 = 121.8$ fF
45°	$L_1 = 329.6$ pH, $L_2 = 960.6$ pH; $C_{\mathrm{off2}} = 225.1$ fF, $C_{\mathrm{off3}} = 263.7$ fF
90°	$L_1 = 795.8$ pH, $L_2 = 397.9$ pH; $C_{\mathrm{off2}} = 318.3$ fF, $C_{\mathrm{off3}} = 636.6$ fF
180°	$L_{\mathrm{LP}} = L_{\mathrm{HP}} = 0.7958$ nH; $C_{\mathrm{LP}} = C_{\mathrm{HP}} = 0.3183$ pF

C. Phase-Cell Synthesis

The 5.625-, 11.25-, and 22.5-degree cells use switched low-pass/band-pass T-networks. The 45- and 90-degree cells use bridged-T variants selected to reduce the difference in phase slope between the two paths. The 180-degree cell uses switched high-pass and low-pass paths that produce approximately opposite 90-degree phase contributions. These architectures follow the general filter-based approach used in prior X-band digital phase shifters [1], [3].

For a phase increment $\Delta\phi$, center angular frequency $\omega_0 = 2\pi f_0$, and reference impedance Z_0, the ideal starting values for the small-bit low-pass/band-pass synthesis are

$$\begin{aligned} L_1 &= \frac{2Z_0}{\omega_0}\tan\left(\frac{\Delta\phi}{2}\right), \quad C_1 = \frac{1}{2\omega_0 Z_0 \tan(\Delta\phi/2)}, \\ L_2 &= \frac{Z_0}{\omega_0}\tan\left(\frac{\Delta\phi}{2}\right), \quad C_2 = \frac{\sin(\Delta\phi)}{\omega_0 Z_0}. \end{aligned} \tag{3}$$

Table I lists the first-order values obtained at $f_0 = 10$ GHz and $Z_0 = 50\ \Omega$. A companion Python calculator regenerates these values and a machine-readable result table. They are analytical starting points, not a claim that the final included schematic retains every value unchanged. Practical tuning compensates for R_{on}, C_{off}, finite inductor quality factor, and loading between cascaded cells.

The 5.625-degree cell illustrates the parasitic sensitivity. Its ideal C_2 is only 31.2 fF, whereas the characterized switch C_{off} is approximately 57.8 fF. Consequently, the switch capacitance is larger than the ideal phase-setting capacitance, so direct application of (3) cannot preserve both phase and matching without circuit-level retuning.

III. SIMULATION DATA AND METRICS

A. Data Acquisition and Integrity

Preserved Cadence exports provide transmission phase, transmission magnitude in decibels, and input reflection magnitude in decibels for every control code at 9, 10, and 11 GHz. The analysis verifies exactly 64 unique states, identical controls across the three files, control voltages restricted to 0 or 1.8 V, and the complete state set $i = 0, \ldots, 63$. The resulting tidy data set contains 192 state-frequency observations. SHA-256 hashes and a Git attribute that disables text normalization protect the raw CSV bytes.

The native CSV exports are treated as authoritative. An audit of the submitted insertion-loss workbook found that its derived 11-GHz column referenced the 10-GHz source column and then repeated that value in the average. The affected workbook is therefore excluded from the reproducible workflow. Recalculation from the native CSV data gives an aggregate mean insertion loss of 15.69 dB rather than the approximately 15.90-dB workbook result.

B. Phase Error

Let $\mathcal{W}(\cdot)$ wrap an angle to $[-180^\circ, 180^\circ)$. The simulated phase of each state is first referenced to state 0 at the same frequency:

$$\begin{aligned} \theta_i(f) &= \mathcal{W}(\angle S_{21,i}(f) - \angle S_{21,0}(f)), \\ e_i(f) &= \mathcal{W}(\theta_i(f) - \phi_{i,\mathrm{ideal}}). \end{aligned} \tag{4}$$

For $N = 64$ states, the common phase offset and offset-removed phase-error root-mean-square (RMS) value are

$$\begin{aligned} \bar{e}(f) &= \frac{1}{N}\sum_{i=0}^{N-1} e_i(f), \\ \sigma_\phi(f) &= \sqrt{\frac{1}{N}\sum_{i=0}^{N-1}[e_i(f) - \bar{e}(f)]^2}. \end{aligned} \tag{5}$$

The total RMS, $\sqrt{N^{-1}\sum_i e_i^2}$, is also reported so that the common offset is not hidden. Aggregate values apply the same population definitions to all 192 observations.

C. Amplitude and Matching

Insertion loss (IL) and input return loss (RL) are derived directly from the simulated scattering parameters:

$$IL_i(f) = -S_{21,i,\mathrm{dB}}(f), \qquad RL_i(f) = -S_{11,i,\mathrm{dB}}(f). \tag{6}$$

Their state variations are population standard deviations about the corresponding mean. Minimum RL is included because a mean value alone can conceal poorly matched control states.

IV. RESULTS

A. Phase Accuracy

At 10 GHz, the common phase offset is -2.17° and the state-dependent RMS error after removing this offset is 2.28°. The corresponding total RMS is 3.15°, demonstrating why the offset definition must be stated. The offset-removed RMS decreases from 3.92° at 9 GHz to 1.55° at 11 GHz, while the mean error changes from -5.03° to $+0.89^\circ$. Therefore, state

TABLE II
RECALCULATED 64-STATE S-PARAMETER STATISTICS

	Phase error (deg)			Insertion loss (dB)			Input return loss (dB)		
f	Mean	Offset-removed RMS	Total RMS	Mean	RMS variation	Min.–max.	Mean	RMS variation	Minimum
9 GHz	−5.03	3.92	6.37	16.26	1.91	12.92–20.62	10.88	3.44	5.92
10 GHz	−2.17	2.28	3.15	15.72	1.66	12.77–19.48	9.46	2.30	5.78
11 GHz	+0.89	1.55	1.79	15.08	1.37	12.59–18.07	8.14	1.70	5.48
Aggregate	−2.10	3.67	4.23	15.69	1.73	12.59–20.62	9.49	2.81	5.48



Fig. 2. Statewise simulation results at 9, 10, and 11 GHz: (a) phase error after subtracting the mean error at each frequency, (b) insertion loss, and (c) input return loss. The dotted horizontal line in panel (c) marks 10 dB.

linearity improves toward 11 GHz even though the common phase bias shifts with frequency.

Across all 192 observations, the global offset-removed RMS is 3.67°. This aggregate value includes both state-dependent dispersion and the variation of the frequency-specific mean errors. It should not be interpreted as a continuous-band metric because only three discrete frequency points are available.

B. Insertion Loss

Mean IL decreases monotonically from 16.26 dB at 9 GHz to 15.08 dB at 11 GHz. The state variation likewise decreases from 1.91 to 1.37 dB. Nevertheless, the absolute loss remains high at every sampled frequency. State 15, corresponding to an ideal phase of 84.375 degrees, produces the maximum IL at all three frequencies: 20.62, 19.48, and 18.07 dB at 9, 10, and 11 GHz, respectively. This persistent extreme indicates a systematic code-dependent loss mechanism rather than random numerical scatter.

The 15.72-dB mean IL obtained at 10 GHz is close in magnitude to the 15.7-dB typical loss reported for a fabricated 0.18-μm CMOS 6-bit X-band design at 9.5 GHz [4]. This numerical similarity is context only: different circuits, models, de-embedding conditions, and the absence of confirmed measurements in the present work prevent a direct performance-equivalence claim.

C. Input Matching

The average input RL degrades with increasing frequency, from 10.88 dB at 9 GHz to 8.14 dB at 11 GHz. More importantly, the minimum RL is below 6 dB at all three frequencies. State 55, corresponding to an ideal phase of 309.375 degrees, is consistently the least-matched state, with RL values of 5.92, 5.78, and 5.48 dB. Hence, even at 9 GHz, where the average RL exceeds 10 dB, the design does not maintain 10-dB matching over all 64 states.

V. CRITICAL ASSESSMENT

A. Dominant Technical Tradeoffs

The simulations demonstrate the intended digital phase progression, but phase accuracy is obtained at the cost of high loss and uneven input matching. Three mechanisms are consistent with the available evidence. First, R_{on} is in series with the RF path and dissipates power in each cascaded stage. Second, finite inductor quality factor adds loss and modifies the ideal network phase. Third, C_{off} is comparable to, or larger than, phase-setting capacitances in the small bits, shifting resonance and disturbing the nominal 50-ohm termination. Cascading six imperfectly matched cells can then magnify state-dependent reflection and transmission variation.

These mechanisms are physically plausible, but the available files do not isolate their individual contributions. In particular, inductor model revisions, substrate loss, layout coupling, and extraction status are not documented sufficiently to support a quantitative loss budget.

B. Evidence and Artifact Boundary

The exact Cadence and simulator versions, TSMC PDK revision, process corner, temperature, and supply-tolerance

conditions are not confirmed. The available files also do not establish whether the 64-state exports are schematic, parasitic-extracted, or electromagnetic post-layout simulations. No verified DRC, LVS, parasitic-extraction, Monte Carlo, chip-area, power-handling, or measured-hardware results are available. Accordingly, this work should be described as a circuit-simulation study, not as a tapeout-ready or experimentally validated integrated circuit.

The public artifact includes the student-created OA schematic source and top-level layout but excludes the licensed TSMC PDK, device models, rule decks, and PDK-derived support files. It also lacks the ADE/Assembler/Maestro state, netlists, and code-sweep definition needed to rerun the original simulations. The preserved outputs and downstream numerical analysis remain available.

C. Recommended Next Steps

The highest-priority work is to establish a traceable simulation baseline: record the PDK and simulator revisions, define process-voltage-temperature corners, and repeat a dense 8–12-GHz sweep. Post-layout parasitic extraction and electromagnetic analysis of inductors, interconnect, and coupling should follow. Circuit optimization should jointly address switch $R_{\mathrm{on}}C_{\mathrm{off}}$ tradeoffs, passive-element quality factor, local matching for the worst cells, and cascade ordering. Finally, the 64-state evaluation should be repeated across corners and Monte Carlo samples before fabrication and measurement.

VI. Conclusion

A 6-bit X-band phase shifter with 64 binary-weighted states was designed in a TSMC 180-nm CMOS environment and evaluated from native Cadence exports at 9, 10, and 11 GHz. The independently recalculated 10-GHz results give a 2.28-degree offset-removed phase-error RMS, 15.72-dB mean insertion loss, and 9.46-dB mean input return loss. Across all 192 samples, the corresponding values are 3.67 degrees, 15.69 dB, and 9.49 dB. The phase response is the strongest aspect of the design; insertion loss, amplitude variation, and worst-state matching are the principal limitations. Further claims require dense-frequency, corner, extracted-layout, electromagnetic, and measured validation.

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